

Companion LX: Weak-Lensing Peak Statistics in the Relaxation-Driven Cyclic Cosmology

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Abstract

Weak-lensing peak statistics probe the nonlinear structure of the gravitational field by identifying local maxima in convergence maps. In the standard cosmological model, the abundance and morphology of peaks are determined by the nonlinear matter distribution, the halo mass function, and the projection of large-scale structure. In the Relaxation-Driven Cyclic Cosmology (RDCC), the scale-dependent evolution of the gravitational potential modifies the depth, coherence, and spatial distribution of peaks in a characteristic way. This Companion develops a continuous description of weak-lensing peak statistics in RDCC, deriving how the relaxation scale influences the peak abundance, the high- χ tail, and the morphology of nonlinear structures. The resulting framework provides a distinctive probe of the RDCC gravitational field beyond two- and three-point statistics.

1 Introduction

Weak-lensing peak statistics identify local maxima in convergence maps and provide a powerful probe of nonlinear structure formation. High peaks correspond to massive halos or line-of-sight projections, while low peaks reflect the interplay of nonlinear structure and large-scale modes.

In the standard cosmological model, the peak abundance and morphology are determined by the nonlinear matter distribution, the halo mass function, and the projection of large-scale structure. In RDCC, the gravitational potential evolves differently. The relaxation mechanism suppresses the decay of small-scale modes and enhances the coherence of large-scale modes. This modifies the depth, coherence, and spatial distribution of peaks in a scale-dependent manner, producing a distinctive signature in weak-lensing peak statistics.

2 The RDCC Convergence Field and Peak Definition

The convergence field is

$$\kappa(\hat{\mathbf{n}}) = \int_0^{\chi_H} d\chi W(\chi) \delta_m(\chi, \hat{\mathbf{n}}). \quad (1)$$

In RDCC, the matter density evolves as

$$\delta_{m,\text{RDCC}}(k, z) = \delta_m(k, z) \left[1 - \chi_\delta \left(\frac{k}{k_{\text{rel}}} \right)^\alpha \right]. \quad (2)$$

The RDCC convergence becomes

$$\kappa_{\text{RDCC}}(\ell) = \kappa(\ell) \left[1 - \chi_\kappa \left(\frac{\ell}{\ell_{\text{rel}}} \right)^\alpha \right]. \quad (3)$$

A peak is defined as a pixel where

$$\kappa(\hat{\mathbf{n}}) > \kappa(\hat{\mathbf{n}} + \Delta\hat{\mathbf{n}}), \quad (4)$$

for all neighboring directions $\Delta\hat{\mathbf{n}}$.

3 Peak Abundance in RDCC

The peak abundance is given by

$$n_{\text{peak}}(\nu) = \frac{dN_{\text{peak}}}{d\nu}, \quad (5)$$

where $\nu = \kappa/\sigma_\kappa$.

In RDCC, the variance becomes

$$\sigma_{\kappa, \text{RDCC}}^2 = \sigma_\kappa^2 \left[1 - \chi_\sigma \left(\frac{\ell}{\ell_{\text{rel}}} \right)^\alpha \right]. \quad (6)$$

The peak abundance becomes

$$n_{\text{peak, RDCC}}(\nu) = n_{\text{peak}}(\nu) \left[1 - \chi_n \left(\frac{\ell}{\ell_{\text{rel}}} \right)^\alpha \right]. \quad (7)$$

This modification suppresses small-scale peaks and enhances large-scale peaks.

4 High-Peak Tail and RDCC Signatures

High peaks correspond to massive halos or strong line-of-sight projections. The high- χ tail is given by

$$n_{\text{peak}}(\nu \gg 1) \propto \exp \left(-\frac{\nu^2}{2} \right). \quad (8)$$

In RDCC, the high- χ tail becomes

$$n_{\text{peak, RDCC}}(\nu \gg 1) \propto \exp \left[-\frac{\nu^2}{2} \left(1 - \chi_{\text{tail}} \left(\frac{\ell}{\ell_{\text{rel}}} \right)^\alpha \right) \right]. \quad (9)$$

This modification reflects the relaxation-driven suppression of small-scale potential wells.

5 Peak Morphology and RDCC Signatures

The relaxation mechanism enhances the coherence of large-scale modes. Peak morphology therefore exhibits:

- broader, smoother peaks at large scales,
- suppressed small-scale substructure,
- a characteristic turnover in the peak-size distribution near ℓ_{rel} ,
- modified correlations between peak height and curvature.

These signatures provide a direct observational probe of the RDCC gravitational field in real-space mass maps.

6 Conclusion

Weak-lensing peak statistics in the Relaxation-Driven Cyclic Cosmology reflect the scale-dependent evolution of the gravitational potential and the nonlinear matter distribution. The relaxation mechanism modifies the peak abundance, the high- χ tail, and the morphology of peaks in a scale-dependent manner, producing distinctive signatures in weak-lensing surveys. These effects provide a direct observational window into the relaxation scale and the temporal structure of the RDCC gravitational field. The present Companion establishes the theoretical foundation for future studies of nonlinear structure, mass mapping, and the interplay between light and gravity in RDCC.

References

- [Jain and Van Waerbeke(2000)] B. Jain and L. Van Waerbeke, *The Astrophysical Journal* **530**, L1 (2000).
- [Kruse and Schneider(2000)] G. Kruse and P. Schneider, *Monthly Notices of the Royal Astronomical Society* **318**, 321 (2000).
- [Fan *et al.*(2010)Fan, Shan, and Liu] Z. Fan, H. Shan, and J. Liu, *The Astrophysical Journal* **719**, 1401 (2010).
- [Liu *et al.*(2015)Liu, Petri, Haiman *et al.*] J. Liu, A. Petri, Z. Haiman, *et al.*, *Physical Review D* **91**, 063507 (2015).
- [Collaboration(2022)] D. Collaboration, *Physical Review D* **105**, 083518 (2022).
- [Collaboration(2021)] K. Collaboration, *Astronomy & Astrophysics* **646**, A140 (2021).
- [Collaboration(2023)] H. Collaboration, *Publications of the Astronomical Society of Japan* **75**, 1 (2023).
- [Kaiser and Squires(1993)] N. Kaiser and G. Squires, *The Astrophysical Journal* **404**, 441 (1993).
- [Jeffrey *et al.*(2018)Jeffrey *et al.*] N. Jeffrey *et al.*, *Monthly Notices of the Royal Astronomical Society* **479**, 2871 (2018).